WORKSHEET - NAPLAN Year 9

Number and Algebra

Fractions, Decimals and Percentages (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-14485

Percy bought 8 packets of cough lollies for $\$18.00$.

The average cost of one packet is

$\$0.45$ $\$2.25$ $\$2.50$ $\$10$

2. Number and Algebra, NAP-79139

A bag of flour weighs $\frac{3}{4}$ of a kilogram.

Peter buys two bags.

How many kilograms of flour does Peter buy?

$\frac{6}{8}$ $\frac{4}{3}$ $1\frac{1}{4}$ $1\frac{1}{2}$

3. Number and Algebra, NAP-96742

Luke had 6 cups of sugar.

He used $\frac{1}{2}$ cup of sugar for one recipe and $2\frac{1}{4}$ cups of sugar for another recipe.

How many cups of sugar did Luke have left?

$2\frac{1}{4}$

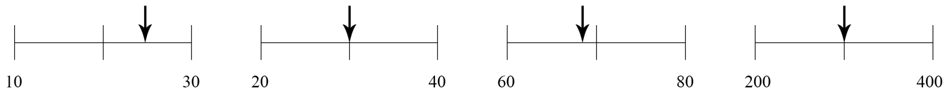
$3\frac{1}{4}$

$3\frac{1}{2}$

$4\frac{3}{4}$

4. Number and Algebra, NAP-26180

Which arrow is pointing to the approximate position of $\sqrt{600}$ on a number line?



5. Number and Algebra, NAP-49729

The table below lists the original price and the amount of discount of a pair of jeans at four different shops.

JEANS SALE		
Shop	Original price	Discount
A	$20	25%
B	$21	$\frac{1}{3}$
C	$18	20%
D	$17	$2 off

Which shop has the lowest sale price for the jeans?

A B C D

6. Number and Algebra, NAP-02689

Ali has a bag of marbles. The marbles are either blue, black or orange.

$\frac{1}{6}$ of her marbles are blue and $\frac{1}{4}$ are black.

What fraction of her marbles are orange?

$\frac{2}{3}$ $\frac{5}{12}$ $\frac{3}{10}$ $\frac{7}{12}$ $\frac{7}{10}$

7. Number and Algebra, NAP-91026

Malcolm will travel by bus for 18 days during November.

A daily ticket will cost him $7.40 and a monthly ticket will cost him $109.80.

What is Malcolm's average **daily saving** if he buys a monthly ticket?

$1.30 $6.10 $23.40 $125.80

8. Number and Algebra, NAP-73312

The original price of a refrigerator is $3200.

Saroo buys the refrigerator at a sale for 20% off the original price.

He has a discount voucher that gives him a further $64 off the sale price.

What percentage of the original price does Saroo pay for the refrigerator?

%

Worked Solutions

1. Number and Algebra, NAP-14485

$$\text{Price of 1 packet} = \frac{\$18.00}{8}$$
$$= \$2.25$$

2. Number and Algebra, NAP-79139

Weight of two bags

$$= 2 \times \frac{3}{4}$$
$$= \frac{6}{4}$$
$$= 1\frac{1}{2} \text{ kilograms}$$

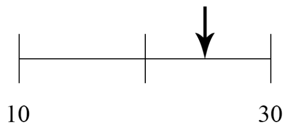
3. Number and Algebra, NAP-96742

$$\text{Cups used} = \frac{1}{2} + 2\frac{1}{4} = 2\frac{3}{4}$$

$$\text{Cups left} = 6 - 2\frac{3}{4} = 3\frac{1}{4}$$

4. Number and Algebra, NAP-26180

$$\sqrt{600} = 24.49...$$



5. Number and Algebra, NAP-49729

Consider the sale price at each shop:

$$A = 20 - (25\% \times 20) = \$15$$
$$B = 21 - \left(\frac{1}{3} \times 21\right) = \$14$$
$$C = 18 - (20\% \times 18) = \$14.40$$
$$D = 17 - 2 = \$15$$

$\therefore$ Shop B has the lowest sale price.

6. Number and Algebra, NAP-02689

Orange marbles

$$= 1 - \left(\frac{1}{6} + \frac{1}{4}\right)$$
$$= 1 - \frac{5}{12}$$
$$= \frac{7}{12}$$

7. Number and Algebra, NAP-91026

Cost of daily tickets

$$= 18 \times 7.40$$
$$= \$133.20$$

Monthly ticket saving

$$= 133.20 - 109.8$$
$$= \$23.40$$

$\therefore$ Average daily saving

$$= \frac{23.40}{18}$$
$$= \$1.30$$

8. Number and Algebra, NAP-73312

$$\begin{aligned}\text{Sale price} &= 80\% \times \$3200 \\ &= \$2560\end{aligned}$$

$$\begin{aligned}\text{Purchase price} &= 2560 - 64 \\ &= \$2496\end{aligned}$$

Percentage of original price

$$\begin{aligned}&= \frac{2496}{3200} \times 100 \\ &= 78\%\end{aligned}$$